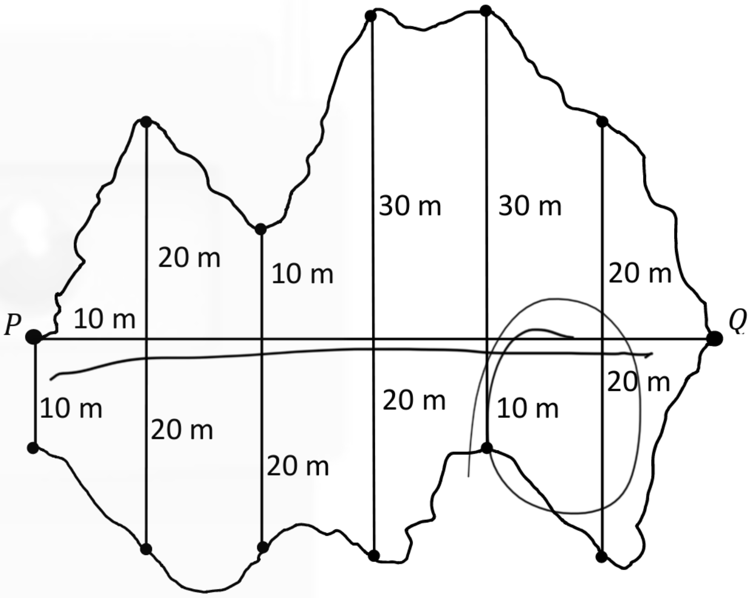
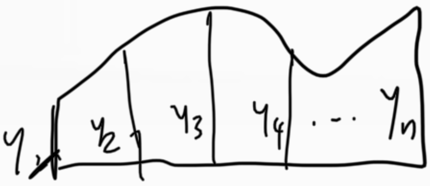
Question 8 **(50 marks)**

The diagram on the right shows the plan of a lake.
The line segment $[PQ]$ represents the distance from the pier, P, to the far side of the lake.
At equal intervals of 10 m along this line, perpendicular measures are made to the sides of the lake as shown.





(a) **(i)** Use the Trapezoidal Rule to estimate the **surface area** of the lake.

$A = \frac{h}{2}(y_1 + y_n + 2(y_2 + y_3 + y_4 + \dots + y_{n-1}))$

$\frac{10}{2}(10 + 0 + 2(40 + 30 + 50 + 40 + 40$

$(10 + 2(200))$

$5(410)$

$2050\ m^2$

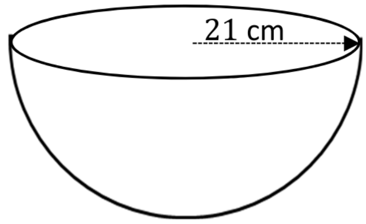
(ii) If the lake is on average 8 m deep, estimate the **volume** of water in the lake.

$2050 \times 8 = 16400\ m^2$

This question continues on the next page.

(b) **(i)** Calculate the **volume** , in cm^3, of a hemisphere of radius 21 cm.
Give your answer terms of π.

$\frac{4}{3}\pi\, 21^3 = 6174\text{cm}^3$



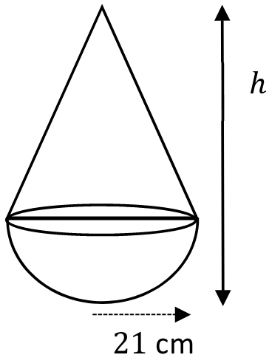
(ii) A buoy on the lake is in the shape of a hemisphere of radius 21 cm surmounted by a cone.
The volume of the cone is equal to the volume of the hemisphere.
Find h, the **total height** of the buoy.

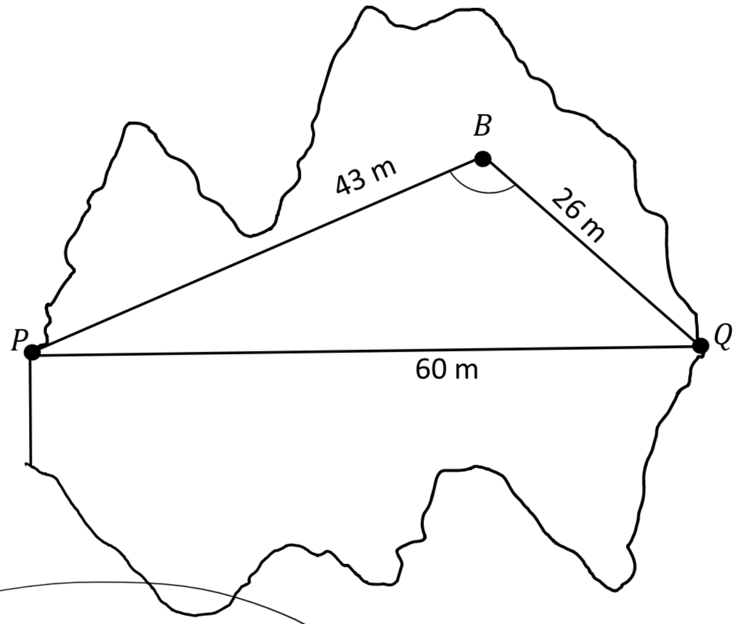
$\frac{1}{3}\pi\, 21^2 h = 6174\text{cm}^3$

$h = \dfrac{6174}{\frac{1}{3}(21)^2}$

$h = 42 + 21$

$= 63\text{cm}$





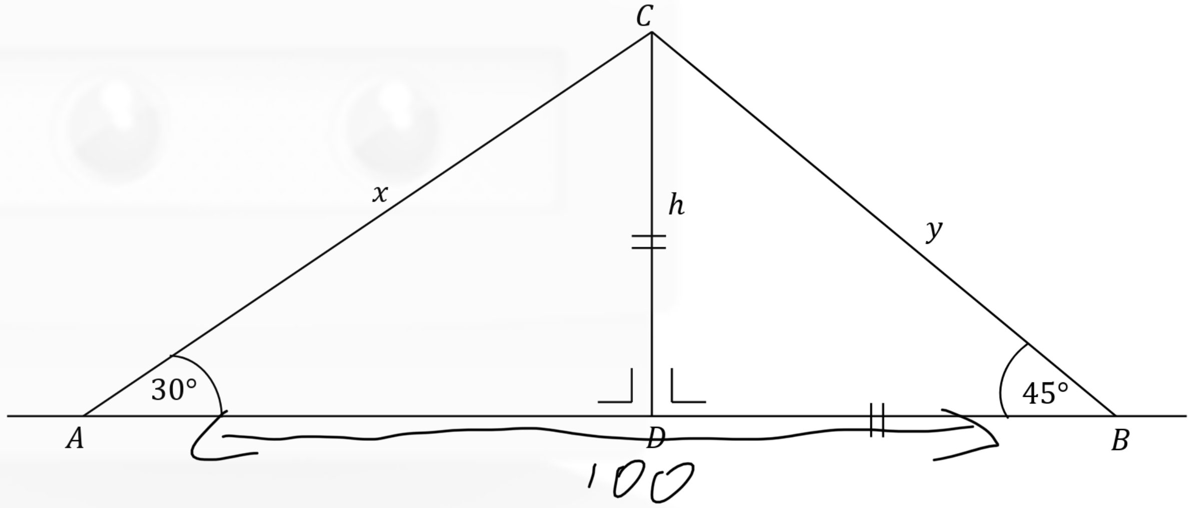
Marking Scheme

(a) (i)	$A = \frac{10}{2}(10 + 0 + 2[200])$ $= 5(410)$ $= 2050\ \text{m}^2$ or $A = \frac{10}{2}\Big(0 + 0 + 2[110]$ $+ \frac{10}{2}(10 + 0 + 2[90])\Big)$ $= 1100 + 950$ $= 2050\ \text{m}^2$	**Scale15D (0, 4, 7, 11, 15)** *Low Partial Credit:* Work of merit • Trapezoidal Rule formula with some substitution *Mid Partial credit:* • Area of one section found *High Partial credit:* Significant Work • Trapezoidal Rule formula with full substitution (for both sections) *Misreading* (−1) Works correctly with Simpson's Rule to Answer $(2233{\cdot}\dot{3}\ \text{m}^2)$
(a) (ii)	$V = 2050 \times 8 = 16400\ \text{m}^3$	**Scale 5B (0, 2, 5)** *Partial Credit:* Work of merit • Some correct substitution into Volume formula *Full Credit:* • Correct answer without supporting work

(b) (i)	$V=\frac{2}{3}\pi(21^3)=6174\pi\ (\text{cm}^3)$	**Scale 5B (0, 2, 5)** *Low Partial Credit:* Work of merit • Writes volume of hemisphere formula • Volume of sphere formula with some • correct substitution • Volume of sphere(12348π cm^3) and stops **Note**: F* applies to: Answer (19396 cm^3) Answer (19386 cm^3) for ($3{\cdot}14$) Answer (19404 cm^3) for $\left(\frac{22}{7}\right)$ *Full Credit:* • Correct answer without supporting work
(b) (ii)	$V=\frac{1}{3}\pi r^2h$ $\frac{1}{3}\pi r^2h=6174\pi$ $h=\dfrac{6174}{\frac{1}{3}r^2}$ or $h=\dfrac{6174}{\frac{1}{3}(21)^2}$ $=42$ $\text{Total Height}=42+21$ $=63\text{ cm}$	**Scale10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Some correct substitution for volume of cone. • Volume of cone = Ans b(i) *Mid Partial Credit:* • Forms equation in h isolated *High Partial Credit:* Significant Work • Height of cone found • One error and finishes correctly

Question 7 **(55 marks)**

A vertical mobile phone mast, $[DC]$, of height h m, is secured with two cables: $[AC]$ of length x m, and $[BC]$ of length y m, as shown in the diagram.
The angle of elevation to the top of the mast from A is 30° and from B is 45°.



(a) **(i)** Explain why $|\angle BCA|$ is 105°.

$|CA| + |CB| = 75°$

$\Rightarrow$ ALL ANGLES IN A TRIANGLE $= 180°$

AND $180° - 75° = 105°$

(ii) The horizontal distance from A to B is 100 m.
Use the triangle ABC to find the length of y.
Give your answer correct to one decimal place.

$|BD| = 50°$ m

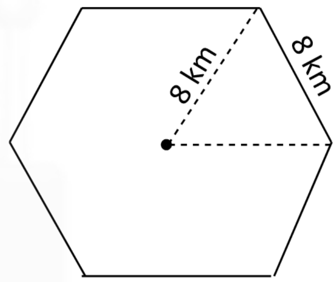
(iii) Using your answer to **Part (a)(ii)** or otherwise, find the value of h and the value of x. Give your answers correct to 1 decimal place.

The height of the mast, h.

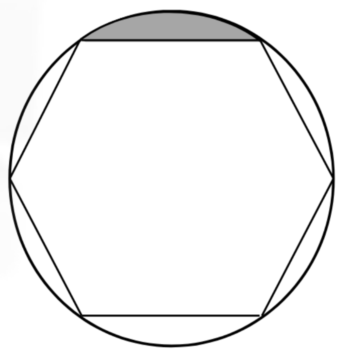
The length of the cable, x.

(b) The two cables to secure the mast costs €25 per metre. The mast itself costs €580 per metre. VAT at 23% is then added in each case.
Calculate the total cost of the cables and mast after VAT is included.

(c) **(i)** The mast can provide a strong signal for an area in the shape of a regular hexagon of side 8 km, as shown in the diagram.
Find the area of the hexagon.
Give your answer in km^2, correct to 2 decimal places.



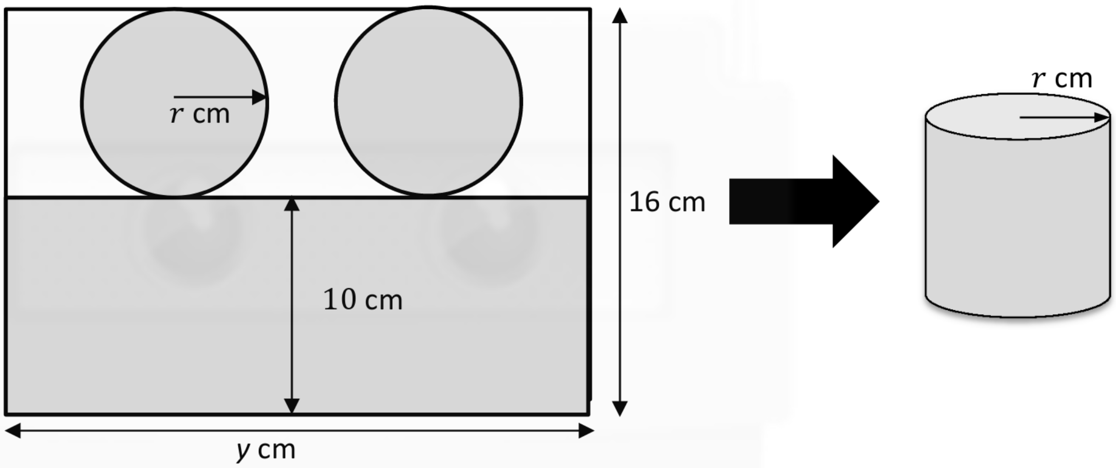
(ii) A circle which touches all vertices of the hexagon can show areas where the signal is weak. One of these areas is shaded in the diagram.
Find this shaded area.
Give your answer in km^2, correct to 1 decimal place.



Q7	Model Solution – 55 Marks	Marking Notes
(a) (i)	$60^\circ + 45^\circ = 105^\circ$ $or\ \ 180^\circ - 30^\circ - 45^\circ = 105^\circ$ $\text{or}\ \ 180^\circ - (30^\circ + 45^\circ) = 105^\circ$	**Scale 10B (0, 5, 10)** *Partial Credit:* Work of merit • Knowledge of sum of angles in a triangle = 180° shown
(a) (ii)	$\frac{\text{y}}{\sin 30} = \frac{100}{\sin 105}$ $\text{y} = \frac{100 \sin 30}{\sin 105}$ $\text{y} = 51{\cdot}7638$ $\text{y} = 51{\cdot}8\text{ m}$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Formula fully substituted
(a) (iii)	$\sin 45 = \frac{h}{51{\cdot}8}$ $\text{h} = 36{\cdot}62813$ $\text{h} = 36{\cdot}6\text{ m}$ $\sin 30 = \frac{36{\cdot}6}{x}$ $x = \frac{36{\cdot}6}{\sin 30}$ $x = 73{\cdot}2\text{ m}$	**Scale 10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *Mid Partial Credit:* • h or x found *High Partial Credit:* • Both formula fully substituted • Incorrect substitution followed by correct solution

(b)	$1{\cdot}23[25(73{\cdot}2 + 51{\cdot}8) + 580(36{\cdot}6)]$ $= 1{\cdot}23(24353)$ $= €29\,954{\cdot}19$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Cost of any element indicated *High Partial Credit:* • Each element excluding VAT formulated
(c) (i)	$\text{Area} = 6\left[\frac{1}{2}(8)(8)\sin 60\right]$ $= 6[27{\cdot}71281292]$ $= 166{\cdot}28\ (\text{km}^2)$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Formula fully substituted • One incorrect substitution followed by correct solution
(c) (ii)	$\text{Area} = \frac{1}{6}[\pi(8^2) - 166 \cdot 28]$ $= \frac{1}{6}[34{\cdot}78192983]$ $= 5{\cdot}7969$ $= 5{\cdot}8\ (\text{km}^2)$	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* Work of merit • Formula with some substitution *High Partial Credit:* • Shaded area fully substituted

A rectangular sheet of aluminium is used to make a cylindrical can of radius r cm and height 10 cm, as shown below. The aluminium does not overlap in the finished can.

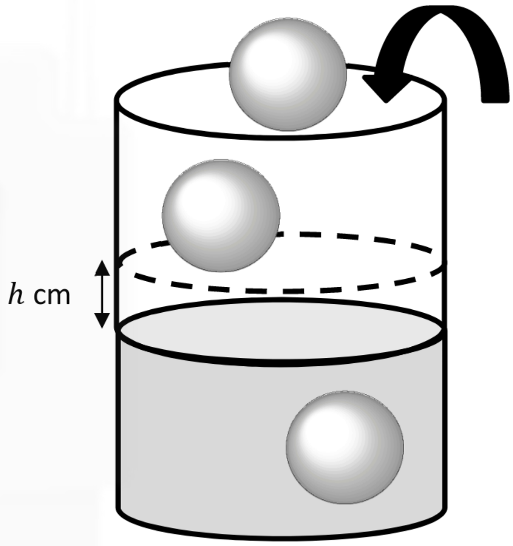


(a) **(i)** Show that r, the radius of the cylinder, is 3 cm.

(ii) Find the distance y. Give your answer correct to the nearest centimetre.

(iii) Find the area, in cm^2, of the waste aluminium after the top, bottom and side of the cylindrical can have been removed from the rectangular sheet.
Give your answer correct to the nearest cm^2.

(b) **(i)** Find the volume of a spherical ice cube of radius 1·5 cm.
Give your answer in terms of π.



(ii) Three of the spherical ice cubes of radius 1·5 cm are added to a cylinder of internal radius 3·5 cm which is partially filled with water.
All of the ice cubes are completely submerged in the water and the water does not overflow.
Find the rise, h cm, in the water level. Give your answer correct to 1 decimal place.

Q9	Model Solution – 40 Marks	Marking Notes
(a) (i)	$10 + Diameter = 16$ $\text{Diameter} = 6$ $\text{Radius} = 3\ (\text{cm})$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Work of merit • Vertical diameter indicated on diagram *High Partial Credit:* • Finds Diameter = 6
(a) (ii)	$y = 2\pi r$ $= 2\pi(3)$ $= 6\pi$ $= 18{\cdot}8495\ \ldots$ $= 19\ (\text{cm})$	**Scale 5B (0, 2, 5)** *Partial Credit* Work of merit • Formula with some substitution *Full Credit:* • Correct answer without supporting work.
(a) (iii)	$\text{Area of sheet} = 6\pi \times 16 = 96\pi$ $\text{C.S.A} = 6\pi \times 10 = 60\pi$ $\text{Area of circles} = 2\pi(3)^2 = 18\pi$ $\text{Waste} = 96\pi - (60\pi + 18\pi)$ $= 96\pi - 78\pi$ $= 18\pi$ $= 56{\cdot}548\ \ldots$ $= 57\ (\text{cm}^2)$	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* • Any area formula with some substitution *High Partial Credit:* • **Two** of the three required areas calculated correctly. • Uses only one circle and finishes consistently

(b) **(i)**	$$\text{Volume} = \frac{4}{3}\pi(1{\cdot}5)^3$$ $$= \frac{4}{3}\pi\left(\frac{27}{8}\right)$$ $$= \frac{9\pi}{2}\ \text{cm}^3$$	**Scale 10B (0, 5, 10)** *Partial Credit:* • Formula with some substitution • Answer not in terms of π $(14{\cdot}137\ \text{cm}^3)$ *Full Credit:* • Correct answer without supporting work
(b) **(ii)**	$$\text{Volume}_{\text{Risen Water}} = \text{Volume}_{\text{Added ice}}$$ $$\pi(3{\cdot}5)^2 h = 3\left(\frac{9\pi}{2}\right)$$ $$h = \frac{27}{2(3{\cdot}5)^2}$$ $$= 1{\cdot}102$$ $$= 1{\cdot}1\ \text{cm}$$	**Scale 10D (0, 3, 5, 8, 10)** *Low Partial Credit:* Work of merit • Formula with some substitution *Mid Partial Credit:* • Both volume formulae fully substituted correctly *High Partial Credit:* • Volume equation set up correctly and correctly substituted • Equation written with h as subject